

Smart City Traffic Intelligence

Part 1: Data Analytics Insights Report

1. Dataset and Data Quality

The Metro Interstate Traffic Volume dataset contains 48,204 records and nine original variables, including traffic volume, temperature, weather conditions and date/time information. Power Query did not identify any missing or error values in the original columns. The value “None” in the holiday column represents non-holiday observations and was therefore treated as a valid category rather than missing data. Further analysis identified repeated timestamps caused by multiple weather observations recorded for the same hour. After considering unique timestamps, the dataset contains approximately 40,575 unique hourly observations. Several temperature values were also recorded close to 0 Kelvin (-273.15°C), which are physically unrealistic and were treated as data-quality outliers in the temperature scatter plot. In addition, some years have incomplete coverage, particularly 2012, 2014 and 2015, so annual comparisons involving these years should be interpreted cautiously.

2. Traffic Trends and Statistical Findings

Traffic volume varied substantially across the observed hours. The mean hourly traffic volume was approximately 3,290.65 vehicles, while the median was 3,427 vehicles. The standard deviation was approximately 1,984.77 vehicles, indicating considerable variation between low-traffic and high-traffic periods. The overall range was 7,280 vehicles. For annual traffic comparisons, 2016 and 2017 provide the most reliable comparison because both years contain full-year data. Total traffic volume increased from approximately 25.03 million in 2016 to 29.42 million in 2017, representing an increase of about 17.53%. The Power BI hourly analysis for 2017 also showed clear variation across the day, with higher traffic concentrated during peak commuting periods rather than remaining constant throughout the day.

3. Weather, Temperature and Congestion Analysis

Temperature showed only a weak positive correlation with traffic volume, with a correlation coefficient of approximately 0.137. This suggests that temperature alone is not a strong predictor of traffic conditions. The probability analysis found that the probability of congestion, defined as traffic volume above 5,500 vehicles, was approximately 14.73%. Clear weather accounted for about 27.78% of observations, while the probability of both congestion and clear weather occurring together was 3.66%. Among congested observations, about 24.83% occurred during clear weather. The odds of congestion under clear weather were also lower than under cloudy conditions, with an odds ratio of approximately 0.735. Power BI analysis further showed that Clouds had the highest average traffic volume at about 3,618.45 vehicles, while

Squall had the lowest at about 2,061.75 vehicles. Overall, weather appears to have some relationship with traffic volume, but it does not explain congestion on its own.

4. Smart City Mobility Recommendations

The analysis suggests that the smart-city mobility team should focus on time-based traffic management rather than relying mainly on weather conditions to predict congestion. Because traffic volume varies strongly by hour, traffic-control resources could be concentrated around recurring peak periods. The city could also use historical traffic, weather, holiday and time-of-day data together in future predictive models, since no single factor appears sufficient on its own. Data quality should also be monitored carefully, particularly repeated timestamps and unrealistic temperature readings, because poor-quality data can reduce the reliability of future AI models and operational decisions. For future development, the dashboard and analytical pipeline could be extended with machine-learning models to forecast congestion and support more proactive traffic management.